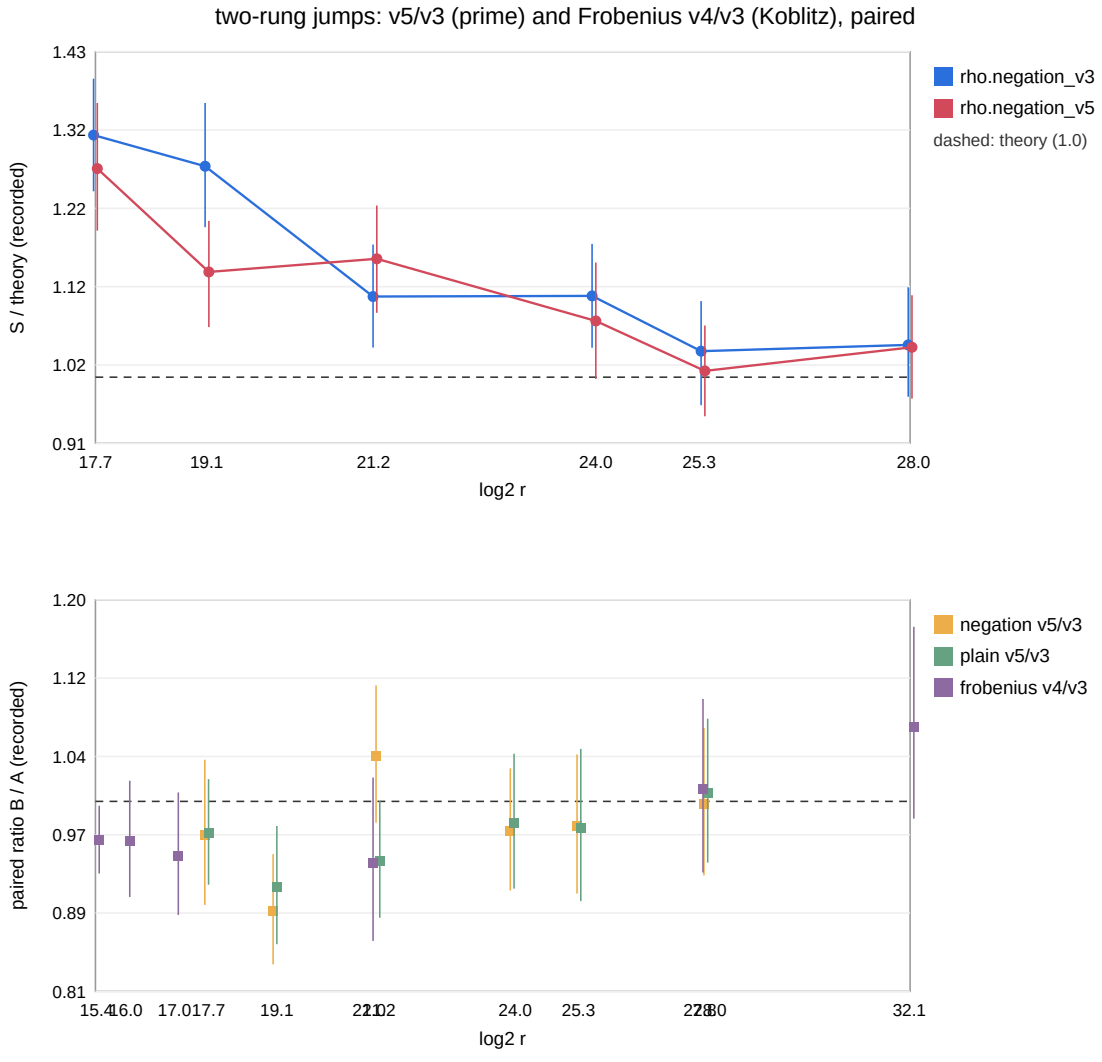
Two-rung jumps: rho v5 and signed-Frobenius v4 against v3

2026-11-07

Two-rung jumps: rho v5 and signed-Frobenius v4 against v3

Result (measured, counted GAE, local L0 host) [pcc:unverified]. Each jump is now the sum of two ladder rungs (one addition) instead of three, with k ≈ √F jumps.

comparison	pairs	paired ratio [95 %]	band (w = 0 / 0.1 / 1)
rho.negation_v5 / rho.negation_v3, prime $2^{17.7–2}28$	2304	**0.972 [0.945, 0.999]**	0.972 / 0.976 / 0.995, order holds
rho.plain_v5 / rho.plain_v3	2304	**0.963 [0.938, 0.988]**	0.963 / 0.964 / 0.978
rho.signed_frobenius_v4 / v3, Koblitz $2^{15.4–2}32.1$	1728	**0.969 [0.943, 0.994]**	0.969 / 0.969 / 0.974



per-size S / theory and paired ratios

Bounds (fit version 2):

bound	id	ops × floor	α
rho.negation_v5, prime	ECBND1h08935fdd1600	1.114	0.468
rho.negation_v3, prime	ECBND1h648847681537	1.147	0.457
rho.signed_frobenius_v4, Koblitz toy	ECBND1h1c9be634d75c	1.813	0.340
rho.signed_frobenius_v3, Koblitz toy	ECBND1h2f3867007530	1.900	0.351

Weighting caveat (from the validator). The pooled ratio weights each curve's S equally. Summed over every run, rho.signed_frobenius_v4 spends **5.5 % more** total GAE than v3 (ratio 1.0546). The two medium curves carry most of the operations, and v4 is not faster there. On the prime session the candidate is 1.045 at 2^21.2, and plain_v5 is 1.008 at 2^28; both intervals include 1.

The Koblitz gain sits on the toy tier: 0.939–0.962 per curve. On the two medium curves the ratios are 1.012 and 1.074 [0.983, 1.173], unresolved, so v4 is not claimed there. rho.negation_v5 has S = 0.921 at 2^28, 1.04× √(π/4).

Validation

An independent validator recomputed the three pooled ratios and their band values to 4.9e-05. It re-verified [d]G = Q on all 13,440 prime and 6,048 Koblitz records, and rejected 240/240 wrong scalars per session (`validation/`). `benchmarks/20261107-rho-v5/VALIDATION.json` covers the prime negation row and reads valid.

Run

item	value
specs	`specs/prime.json`, `specs/koblitz.json`
sessions	prime `ECBS1h288d047c6183` (13,440 records), koblitz `ECBS1h882e6d561549` (6,048); all verified
binary	sha256 `9fde9d8babe2…`, built from `8b8d30884f` (source of the v5 ids)
audits	`audit-prime.json`: replay-all 11,520; `audit-koblitz.json`: 5,184; 0 problems
A/A	ops 1.0000 in both sessions
isolation	L0 container